



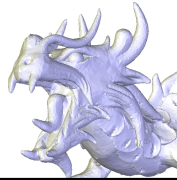
Lyon 1

Meshes and Computational Geometry

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M2 ID3D
Image, Développement
et Technologie 3D
et 3A Centrale

2026-27



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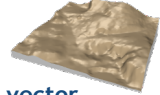
Local surface variations

- How to study and characterize surface variations?

- When defined as a height function over a plane?

$$z=f(u,v)$$

Multivariate scalar function



- When parameterized as a 3D vector function?

$$\mathbf{p}(u,v) = \begin{pmatrix} x(u,v) \\ y(u,v) \\ z(u,v) \end{pmatrix}, (u,v) \in \mathbb{R}^2$$

$$\Omega \subset \mathbb{R}^2$$

$$\xrightarrow{P}$$

$$\xleftarrow{U}$$

$$\mathcal{S} \subset \mathbb{R}^3$$

$$45$$

$$Boundary$$

$$Boundary$$

- When no parameterization is provided?

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Interest

- Local surface variations
 - Used to compute geometric information such as normal vectors, curvature, and higher order information ...
- Useful for :
 - Surface analysis
 - Surface rendering
 - Surface texturing
 - Constructing a better parameterization with less distortion

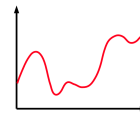
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Differential operators

- Reminder :

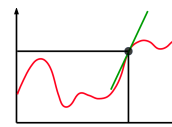
- Given a univariate function $f : \mathbb{R} \rightarrow \mathbb{R}$



- The derivative of f is another function $\frac{\partial f}{\partial x}$ that describes the growing speed of f

$$f' : \mathbb{R} \rightarrow \mathbb{R}$$

$$f'(x) = \lim_{a \rightarrow 0} \frac{f(x+a) - f(x)}{a}$$

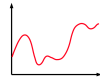


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Differential operators

- Reminder :

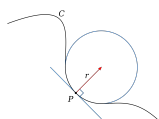
- The Laplacian (second derivative) of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is a measure of the difference between the value of f at any point P and the average value of f in P's neighborhood



- It is linked to the curvature of the curve (inverse of the osculating circle radius)

$$\kappa(x) = \frac{|f''(x)|}{(1 + (f'(x))^2)^{3/2}}$$

Since the curve is oriented, the norm $|\cdot|$ can be removed to get a signed curvature



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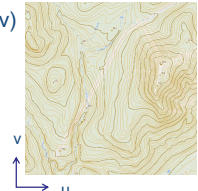
Differential operators

- Derivatives of a multivariate functions :

- Gradient of a scalar function $z=f(u,v)$

$$\text{grad}(f) = \begin{bmatrix} \frac{\partial f}{\partial u} \\ \frac{\partial f}{\partial v} \end{bmatrix}$$

- Orthogonal to iso-lines
- Scalar product with a direction U,V gives the derivative in direction U,V



- Laplacian of a scalar function $f(u,v)$

$$\Delta(f) = \frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2}$$

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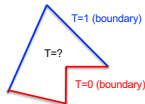
Differential operators

- Laplace equation (useful for interpolation)



$$f : U \in \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\Delta f = 0$$



- Solutions of Laplace equation :

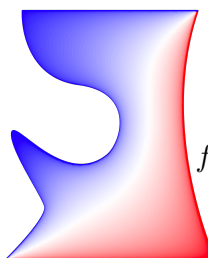
- Harmonic functions = kernel of the Laplacian operator
- No local maxima or minima in U
- Minimizing the Dirichlet energy that measures the smoothness of a function

$$E(f) = \int_U \langle \nabla f, \nabla f \rangle dA$$

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Differential operators

- Solutions of Laplace equation :
 - Used for interpolation



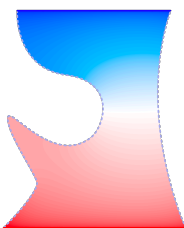
[K. Crane]

$$\begin{aligned} \Delta f(x) &= 0 & x \in U \\ f(x) &= f_0(x) & x \in \delta U \end{aligned}$$

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Differential operators

- Solutions of Laplace equation :
 - Used for interpolation



$$\Delta f(x) = 0 \quad x \in U$$

$$f(x) = f_0(x) \quad x \in \delta U_D \text{ (Dirichlet boundary)}$$

$$\nabla f \cdot n = g_0(x) \quad x \in \delta U_N \text{ (Neumann boundary)}$$

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Differential operators

- Harmonic functions not to be misunderstood with the eigenvectors and eigen values of the Laplacian operator

$$\Delta f = \lambda f$$

When $U = \mathbb{R}$ eigenvectors are the sines and cosines functions that are used within Fourier framework

$$f''(x) = \lambda f$$

$$f(x) = -\cos(\lambda x + cste)$$

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Differential operators

- In a multivariate context, differential operators are often used in differential equations

- Generally expressed in $\mathbb{R}^k$

- Spatial notation "nabla" :
Think of it as a vector
of partial derivative operators!

$$\nabla = \begin{bmatrix} \delta_u \\ \delta_v \end{bmatrix}$$

- Gradient operator of a **scalar** function f : ∇f
- Divergence operator of a **vector** function $\mathbf{V}$: $\nabla \cdot \mathbf{V}$
- Laplacian operator of a **scalar** function f : $\Delta f = \nabla \cdot \nabla f$
- Curl (rotational) operator of a **vector** function $\mathbf{V}$: $\nabla \times \mathbf{V}$

- Expression of nabla depending on the coordinate system of the input domain

Cartesian
coordinates u,v

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Gradients, divergence et rotationnels

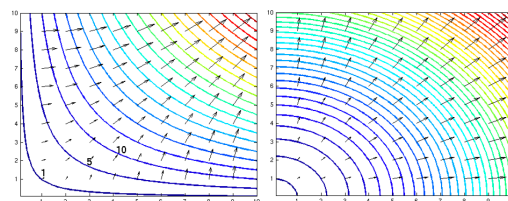


FIGURE 1 – Exemples de champs scalaires (couleur, valeurs élevées en rouge) et de leur gradient (flèches). À gauche : $f(x,y) = xy$, à droite $f(x,y) = x^2 + y^2$.

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Gradients, divergence et rotationnels

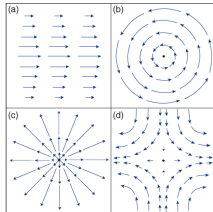


FIGURE 2 – Exemples de champs de vecteurs bidimensionnels : (a,b) champs rotationnels ; cisaillement pur (a) et rotation solide (b) (imaginer la rotation d'une petite roue à aubes insérée dans l'écoulement). (c) champ central divergent. (d) champ avec déformation, mais à divergence et rotationnel nuls (compression dans une direction, dilatation dans l'autre à surface constante).

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Expression of nabla on different coordinates systems

Opération	Coordonnées cartésiennes (x, y, z)	Coordonnées cylindriques (r, θ, z)	Coordonnées sphériques (r, θ, φ)
Définition des coordonnées		$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$	$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$
$\vec{A}$	$A_x \vec{u}_x + A_y \vec{u}_y + A_z \vec{u}_z$	$A_r \vec{u}_r + A_\theta \vec{u}_\theta + A_z \vec{u}_z$	$A_r \vec{u}_r + A_\theta \vec{u}_\theta + A_\varphi \vec{u}_\varphi$
$\vec{\nabla}$	$\frac{\partial}{\partial x} \vec{u}_x + \frac{\partial}{\partial y} \vec{u}_y + \frac{\partial}{\partial z} \vec{u}_z$	$\frac{\partial}{\partial r} \vec{u}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \vec{u}_\theta + \frac{\partial}{\partial z} \vec{u}_z$	$\frac{\partial}{\partial r} \vec{u}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \vec{u}_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \varphi} \vec{u}_\varphi$

[Wikipedia]

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Differential operator

- A few reminders :
 - Gradient** : direction $(u_{\max}, v_{\max})$ of maximal slope
 - Divergence** : flow traversing a unit element around (u, v)
 - Laplacian** : measures the difference between the function and its mean value in a small neighborhood
 - Useful for several geometry processing tasks (interpolation by heat diffusion, spectral analysis, mean-curvature, smoothing)
 - Curl** (Rotational) : does the vector field locally turn around one vector?
- Questions :
 - How could you define the "gradient" of a vector field? Jacobian matrix
 - How could you define the "laplacian" of a vector field? Vector with the Laplacian of each component

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Other use of Nabla

- Nabla transpose
 $x \delta$: Derivative of the scalar field on the left
 $\nabla^t = [x \delta, y \delta, z \delta]$
- Jacobian matrix of a vector field $\mathbf{V}$

$$\mathbb{J}_{\mathbf{V}} = \mathbf{V} \nabla^t = \begin{bmatrix} \delta_x v_x, \delta_y v_x, \delta_z v_x \\ \delta_x v_y, \delta_y v_y, \delta_z v_y \\ \delta_x v_z, \delta_y v_z, \delta_z v_z \end{bmatrix}$$

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Differential operator

- Functions defining a curve
 $\mathbf{x} : [0, L] \rightarrow \mathbb{R}^3$ de classe C^3
 Tangent, normal and binormal vectors?
- Tangent vector : $\mathbf{t}(t) = \frac{\mathbf{x}'(t)}{\|\mathbf{x}'(t)\|}$ $\mathbf{x}'(t)$ unitary if $\mathbf{x}(t)$ isometric
- Normal vector : $\mathbf{m}(t) = \frac{\mathbf{t}'(t)}{\|\mathbf{t}'(t)\|}$ Unitary vector derivatives
- Binormal vector : $\mathbf{b}(t) = \mathbf{t}(t) \times \mathbf{m}(t)$
- Serret-Frénet frame $(\mathbf{x}(t); \mathbf{t}(t), \mathbf{m}(t), \mathbf{b}(t))$

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Differential operator

- Let s be the curvilinear abscissa
 - s : length of curve between $\mathbf{x}(0)$ and $\mathbf{x}(t)$
 $s(t) = \int_0^t \|\mathbf{x}'(t)\| dt$
 - parameterize the curve wrt s
 - $\mathbf{t}(s) = \mathbf{x}'(s)$ is a unit vector
- Courbure** : $\kappa(s) = \mathbf{x}''(s)$
 - Mesure la déviation par rapport à une droite
- Torsion** : $\tau(s) = \frac{\det[\mathbf{x}'(s), \mathbf{x}''(s), \mathbf{x}'''(s)]}{\kappa^2(s)}$
 - Mesure le défaut de planarité

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Differential operator

- Functions defining a surface

$$\mathbf{X} : \Omega \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3 \text{ de classe } C^r$$

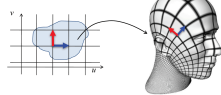
Analogues vecteurs tangent/normal/binormal ?

On note $\mathbf{X}_u = \frac{\partial \mathbf{X}}{\partial u}$ et $\mathbf{X}_v = \frac{\partial \mathbf{X}}{\partial v}$

Plan tangent en p : $T_p \mathbf{X} = \text{plan passant par } p \text{ et engendré par les vecteurs } \mathbf{X}_u(p) \text{ et } \mathbf{X}_v(p)$

Vecteur normal : $\mathbf{n}(p) = \frac{\mathbf{X}_u(p) \times \mathbf{X}_v(p)}{\|\mathbf{X}_u(p) \times \mathbf{X}_v(p)\|}$

$(\mathbf{X}(p); \mathbf{X}_u(p), \mathbf{X}_v(p), \mathbf{n}(p))$ forme aussi un repère local



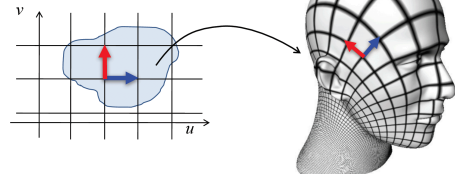
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Differential operator

- Functions defining a surface
- How do unitary vectors u, v are transformed by the parameterization?
 - Length and angles distortion?

$$\mathbf{X} : \Omega \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3 \text{ de classe } C^r$$



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Differential operator

- Functions defining a surface?

On note de manière analogue $\mathbf{X}_{uu} = \frac{\partial^2 \mathbf{X}}{\partial u^2}$, etc.

Première forme fondamentale :

$$\mathbf{I} = \begin{bmatrix} E & F \\ F & G \end{bmatrix} := \begin{bmatrix} \mathbf{X}_u^T \mathbf{X}_u & \mathbf{X}_u^T \mathbf{X}_v \\ \mathbf{X}_v^T \mathbf{X}_u & \mathbf{X}_v^T \mathbf{X}_v \end{bmatrix}$$

Angle change
To compute the dot product between two tangent vectors
Length change

Seconde forme fondamentale :

$$\mathbf{II} = \begin{bmatrix} e & f \\ f & g \end{bmatrix} := \begin{bmatrix} \mathbf{X}_{uu}^T \mathbf{n} & \mathbf{X}_{uv}^T \mathbf{n} \\ \mathbf{X}_{uv}^T \mathbf{n} & \mathbf{X}_{vv}^T \mathbf{n} \end{bmatrix}$$

Squared distance from the surface to the tangent plane for a displacement u, v

Opérateur de forme/Application de Weingarten :

$$\mathbf{W} := \frac{1}{EG - F^2} \begin{bmatrix} eG - fF & fG - gF \\ fE - eF & gE - fF \end{bmatrix} = (D_u \mathbf{n} \ D_v \mathbf{n})$$

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$\mathbf{I}$ = outil géométrique (**tenseur métrique**)

- ▶ Permet de mesurer aires locales, longueurs de courbes sur la surface, angles, ...

$$dA = \sqrt{EG - F^2} du dv$$

- ▶ Exemple : anisotropie locale de la surface : décomposition spectrale de $\mathbf{I}$

Propriétés différentielles ne dépendant que de $\mathbf{I}$ sont dites **intrinsèques**

- ▶ Ne dépendent pas de la paramétrisation
- ▶ Ne dépendent pas de l'espace 3D

Eigenvalues of $\mathbf{I}$: maximal/minimal stretching of a tangent vector

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$\mathbf{II}$ = propriétés **extrinsèques** de la surface

- ▶ Dépendent du plongement dans l'espace ambiant $\mathbb{R}^3$

$\mathbf{W}$ détermine les directions de courbure locale de la surface

- ▶ Valeurs propres = courbures principales
- ▶ Vecteurs propres = directions principales de courbure

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Courbures principales et directions principales de courbure :

$$\mathbf{W} = \begin{bmatrix} \bar{\mathbf{t}}_1 & \bar{\mathbf{t}}_2 \end{bmatrix} \begin{bmatrix} \kappa_1 & 0 \\ 0 & \kappa_2 \end{bmatrix} \begin{bmatrix} \bar{\mathbf{t}}_1 & \bar{\mathbf{t}}_2 \end{bmatrix}^{-1}$$

Courbure moyenne : $H = \frac{\kappa_1 + \kappa_2}{2} = \frac{1}{2} \text{trace}(\mathbf{W})$

Courbure de Gauss : $K = \kappa_1 \cdot \kappa_2 = \det(\mathbf{W})$

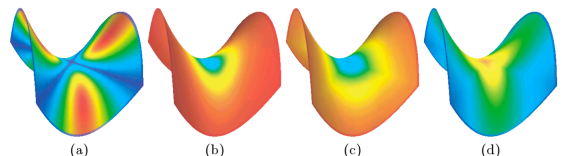


Fig. 5. Curvature plots of a triangulated saddle using pseudo-colors: (a) Mean, (b) Gaussian, (c) Minimum, (d) Maximum.

[Meyer et al. 2003]

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Implicit surfaces

2.1.2. Differential Operators for implicit Surfaces
Assume that

$$\Gamma_c := \{x = [x_1, x_2, x_3]^T \in \mathbb{R}^3 : \phi(x) = c\}$$

where $\phi(x)$ is a properly smooth function defined on $\mathbb{R}^3$, c is an arbitrarily given constant. Suppose that $\|\nabla\phi\|$ on Γ_c , thus, according to the implicit function theorem, the level-set surface could be locally parameterized. For each point on the surface, we can obtain the unit normal vector

$$n = \frac{\nabla\phi}{\|\nabla\phi\|}.$$

The mean curvature H for the level-set surface can be deduced as

$$H = -\frac{1}{2} \operatorname{div} \left(\frac{\nabla\phi}{\|\nabla\phi\|} \right) \quad (3)$$

where ∇ and div denote the classical gradient and divergence operators, respectively.

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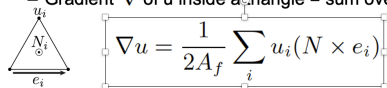
Discrete differential operators on a surface

- Functions defined on a surface
 - Let u be a scalar function defined on a surface
 - We would like to express local variations of u
- Let consider the discrete case of a surface being approximated by a simplicial mesh
 - **Function** u discretized on vertices
 - **Gradient** of u discretized on triangles
 - **Divergence** of a vector (defined on triangles) discretized on vertices
- Bibliography : Keenan Crane

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Discrete gradient on a mesh

- Gradient of a function u defined on the vertices of a triangle
 - Gradient of U the linear interpolation of u
 - U is the sum of 3 functions $U = U_0 + U_1 + U_2$
 - With U_i the affine function equal to u_i at vertex i and zero on the opposite edge e_i
 - Gradient of U_i orthogonal to e_i (with amplitude u_i/h_i where h_i height of the triangle ($h_i = 2A_f/|e_i|$))
 - Gradient ∇ of u inside a triangle = sum over the 3 vertices



$$\nabla u = \frac{1}{2A_f} \sum_i u_i (N \times e_i)$$

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Discrete Dirichlet energy

- Dirichlet energy $E(u_{\text{Triangle}})$ of the function u restrained to a single triangle
 - Integral of the squared norm of the gradient (scalar quantity)

$$E(u_{\text{Triangle}}) = \frac{1}{2} \int_{\text{Triangle}} \|\nabla(u)\|^2 ds$$

$$E(u_{\text{Triangle}}) = \frac{1}{4A_f} \sum_{j=0..3} \cot(\alpha_j) (u_{(j+1)\%3} - u_{(j+2)\%3})$$

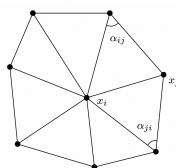
- Where α_j is the angle at the j^{th} vertex of the face
- A_f is the area of the triangle



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Discrete Laplace Beltrami on a mesh

- Laplacian at a vertex of a function u defined on the mesh vertices
 - Difference between the value of u at the i^{th} vertex and the mean at neighbor vertices
 - Which weights should be used for the mean?



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Discrete Laplace Beltrami on a mesh

- Laplacian at a vertex of a function u defined on the mesh vertices
 - Difference between the value of u at the vertex and the mean at neighbor vertices
 - Which weights should be used for the mean?
- Properties :
 - The Laplacian of u at vertex i is the **derivative** of the Dirichlet energy of u on the whole mesh **wrt the value** u_i

$$\Delta u_i = \frac{\delta E(u_{\text{Mesh}})}{\delta u_i}$$

where

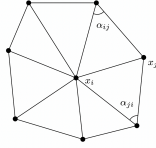
$$E(u_{\text{Mesh}}) = \frac{1}{4} \sum_{f=0}^{f=n_f-1} \frac{1}{A_f} \sum_{j=0}^{j=2} \cot(\alpha_{(f,j)}) (u_{(f,(j+1)\%3)} - u_{(f,(j+2)\%3)})$$

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Discrete Laplace Beltrami on a mesh

Around vertex i there are two faces sharing the edge between vertex i and the neighbor vertex j

$$\Delta u_i = \frac{\delta E(u_{Mesh})}{\delta u_i}$$



$$\Delta u_i = \frac{1}{2A_i} \sum_{j \in neighbor(i)} (\cot \alpha_{ij} + \cot \alpha_{ji})(u_j - u_i)$$

Where A_i is the surface area around vertex i

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Discrete Laplace Beltrami on a mesh

- Another interpretation of the discrete Laplacian cotan formula
 - Laplacian at vertex i of the function u defined on the mesh vertices
 - Difference between the values of u at vertex i and the mean at neighbor vertices
 - The weights used for the mean must be consistent with the fact that the Laplacian of the position vector (ie. 3 Laplacian for x , y and z respectively) at vertex i must correspond to the gradient of the mesh surface area when vertex i is moving
 - The Laplacian of the position vector is a vector oriented in the normal direction (or in the opposite direction depending on the convexity), with amplitude the mean curvature at vertex i

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Differential geometry

- How to generalize normal, curvatures on a mesh?
- Consider u corresponding to each coordinate function in turn
 - $u = {}^t(x, y, z)$

$$\Delta \mathbf{x} u = -2H \mathbf{n}$$

H : mean curvature

$\mathbf{n}$: normal vector

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Differential operators

- Simplicial meshes
 - Laplacian Δ of u at vertex i = sum over neighbor vertices j

$$(Lu)_i = \frac{1}{2A_i} \sum_j (\cot \alpha_{ij} + \cot \beta_{ij})(u_j - u_i)$$

- Gradient ∇ of u inside a triangle = sum over the 3 vertices i

$$\nabla u = \frac{1}{2A_f} \sum_i u_i (N \times e_i)$$

- Divergence at vertex i of a vector $\mathbf{X}$ defined on faces = sum over incident faces j

$$2A_f \nabla \cdot \mathbf{X} = \frac{1}{2} \sum_j \cot \theta_1 (e_1 \cdot \mathbf{X}_j) + \cot \theta_2 (e_2 \cdot \mathbf{X}_j)$$

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Area A_i

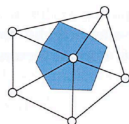
- Computed by duality to a vertex



Barycentric cell



Voronoi cell



Mixed Voronoi cell

Courtesy M. Botsch et al.

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Can be expressed with vectors and matrices

- Function u described as a vector $\mathbf{U}$ of size nv (number of vertices)
- Laplacian described as a $nv \times nv$ matrix

$$L_{ii} = -\frac{1}{2A_i} \sum_{neighbor\ j} (\cot \alpha_{ij} + \cot \alpha_{ji}) \text{ Diagonal}$$

$$L_{ij} = \frac{1}{2A_i} (\cot \alpha_{ij} + \cot \alpha_{ji}) \text{ if } i \text{ and } j \text{ neighbours}$$

$$L_{ij} = 0 \text{ otherwise}$$

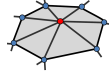
- The product $\mathbf{LU}$ gives a vector of the nv discrete Laplacian of $\mathbf{U}$

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Using the Laplacian operator for parameterizing a shape

- Harmonic Mapping :
 - The 2D positions $\mathbf{U} = (u, v)$ are to be defined at each vertex
 - Modify the vertices 3D positions to flatten the mesh (decrease the curvature to 0 **without changing the weights in the Laplace operator**)
 - Laplace operator (defined on the 3D mesh) of the 2D coordinates gives zero

$$\Delta \mathbf{U} = 0$$

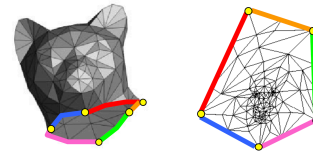


- Boundary points are previously mapped and fixed on a circle (Convex mapping)

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Convex mapping

- Boundary vertices are fixed (Tutte, Floater)
- The weights should be positive to avoid triangles fold-over

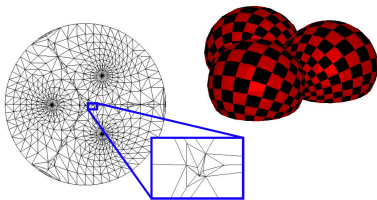


$\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_n$ - inner vertices
 $\mathbf{V}_{n+1}, \dots, \mathbf{V}_N$ - boundary vertices

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Convex mapping

- Boundary vertices are fixed (Tutte, Floater)
- The weights should be positive.



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Signals being studied in Computer Graphics

- Signals defined on $\mathbb{R}^2$
 - Scalar signals :
 - height value (terrain)
 - density of some fluid flowing in the plane
 - Vector fields
 - Surface parameterization $\Omega \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$
 - Displacement field of some fluid flowing in the plane
- Signals defined in $\mathbb{R}^3$
 - Density of a volume material (scalar)
 - Displacement field of some fluid (vector)

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Signals being studied in Computer Graphics

- Signals defined on a surface $\mathbb{S}$
 - Scalar values :
 - Temperature, grey color ...
 - Position coordinates (x, y or z)
 - Vector fields
 - Normal vector
 - Maximal/minimal curvature direction
 - Displacement field
 - ...

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